

UT10 Compulsory task - MongoDB

Nombre: Mark Pires Guimaraes

Activitat 1

He creado la base de datos **sample_supplies** con la colección **sales**.

Activitat 2

2.1

```
> db.sales.countDocuments({})  
< 5000  
sample_supplies> |
```

2.2

```
> db.sales.countDocuments({ storeLocation: { $in: ["London", "New York"] } })  
< 1295
```

2.3

```
> db.sales.countDocuments({  
  $or: [  
    { storeLocation: "London" },  
    { storeLocation: "New York" }  
  ]  
})  
< 1295
```

2.4

```
> db.sales.countDocuments({  
  storeLocation: { $in: ["Denver", "London"] },  
  purchaseMethod: "Phone"  
})
```

< 275

2.5

```
> db.sales.countDocuments({
  storeLocation: "New York",
  items: { $size: 1 },
  "items.0.quantity": 1
})
< 4
```

2.6

```
> db.sales.countDocuments({
  items: {
    $elemMatch: {
      name: "laptop",
      price: { $gte: 900, $lte: 950 }
    }
  }
})
< 100
```

2.7

```
> db.sales.countDocuments(
  {storeLocation: "New York",
  items: {
    $elemMatch: {
      name: "laptop",
      price: { $gte: 900, $lte: 950 }
    }
  }
})
< 11
```

2.8

```
> db.sales.find(
  { "items.price": { $gt: 1598 } },
  {
    _id: 0,
    "items.name": 1,
    "items.price": 1,
    "items.quantity": 1,
    storeLocation: 1,
    "customer.email": 1
  }
).sort({ "customer.email": 1 })
< {
```

2.9

```
> db.sales.find(
  { "items.name": "laptop" },
  { _id: 0, storeLocation: 1, "customer.email": 1 }
).sort({ "items.price": -1 }
)
.limit(5)
```

2.10

```
> db.sales.insertOne(
  {saleDate: ISODate("2026-05-28"),
   items: [
     {
       name: "laptop",
       tags: ["electronics"],
       price: 1100.50,
       quantity: 1
     }
   ],
   storeLocation: "Barcelona",
   customer: {
     gender: "F",
     age: 28,
     email: "marta@test.com"
   },
   couponUsed: false,
   purchaseMethod: "In store"
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a187080bf2b4ac8fe8484f1')
}
```

2.11

```
> db.sales.find(
  { "customer.email": "marta@test.com" }
)
< {
  _id: ObjectId('6a187080bf2b4ac8fe8484f1'),
  saleDate: 2026-05-28T00:00:00.000Z,
  items: [
    {
      name: 'laptop',
      tags: [
        'electronics'
      ],
      price: 1100.5,
      quantity: 1
    }
  ],
  storeLocation: 'Barcelona',
  customer: {
    gender: 'F',
    age: 28,
    email: 'marta@test.com'
  },
  couponUsed: false,
  purchaseMethod: 'In store'
}
```

2.12

```
> db.sales.updateOne(
  { "customer.email": "marta@test.com" },
  {
    $push: {
      items: {
        name: "mouse",
        tags: ["accessory"],
        price: 25.00,
        quantity: 1 }}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

La consulta:

```
> db.sales.find({ "customer.email": "marta@test.com" })
< {
  _id: ObjectId('6a187080bf2b4ac8fe8484f1'),
  saleDate: 2026-05-28T00:00:00.000Z,
  items: [
    {
      name: 'laptop',
      tags: [
        'electronics'
      ],
      price: 1100.5,
      quantity: 1
    },
    {
      name: 'mouse',
      tags: [
        'accessory'
      ],
      price: 25,
      quantity: 1
    }
  ],
  storeLocation: 'Barcelona',
  customer: {
    gender: 'F',
    age: 28,
    email: 'marta@test.com'
  },
}
```


2.13

```
> db.sales.deleteOne({ "customer.email": "marta@test.com" })
< {
  acknowledged: true,
  deletedCount: 1
}
```

La consulta:

```
> db.sales.find(
  { "customer.email": "marta@test.com" }
)
```